

INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



Assignment Report

M.Tech in Integrated Circuit Design and Technology (ICDT)

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Subject Name:	CMOS Analog IC Design
Subject Code:	EE 651
Assignment No:	04
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Question 1

Design a Cascode Amplifier with active load for the following specifications:

- Voltage Gain > 60 dB
- Output swing > 650 mV
- Power budget $< 20 \mu\text{W}$ (only for cascode circuit)
- Load capacitance = 1 pF

Calculate the bandwidth (BW) and unity gain frequency of your designed amplifier. Also, write the gain margin and phase margin.

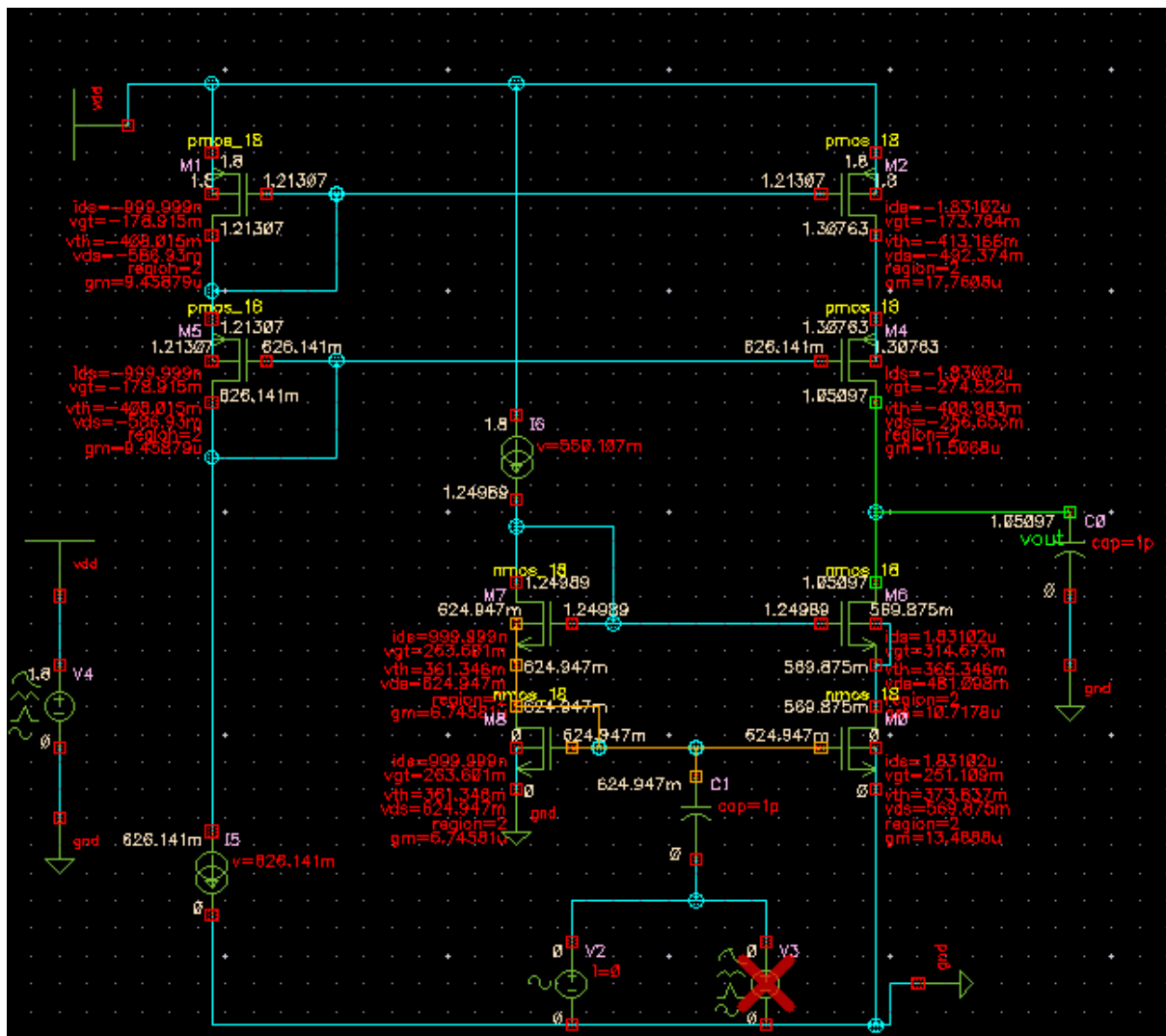


Figure 1: Schematic

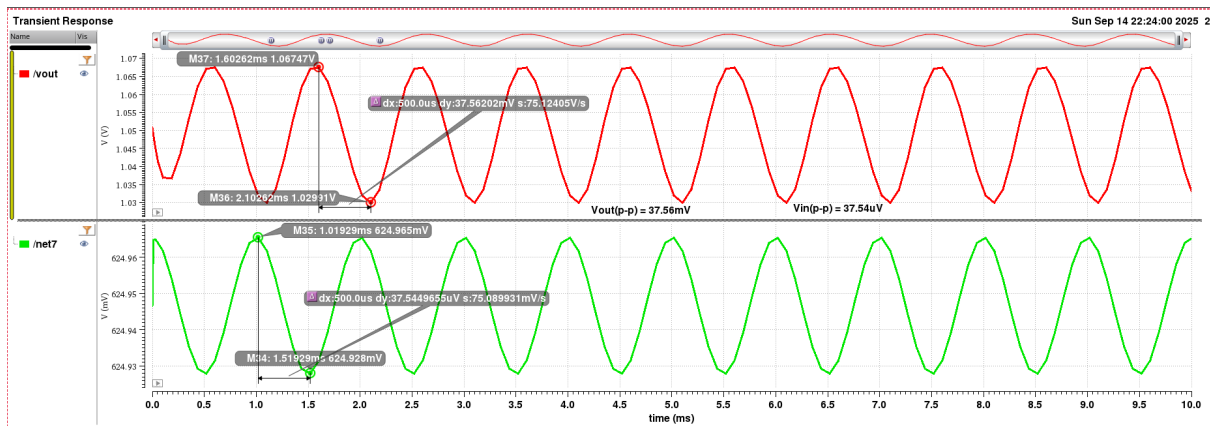


Figure 2: Trasient Analysis

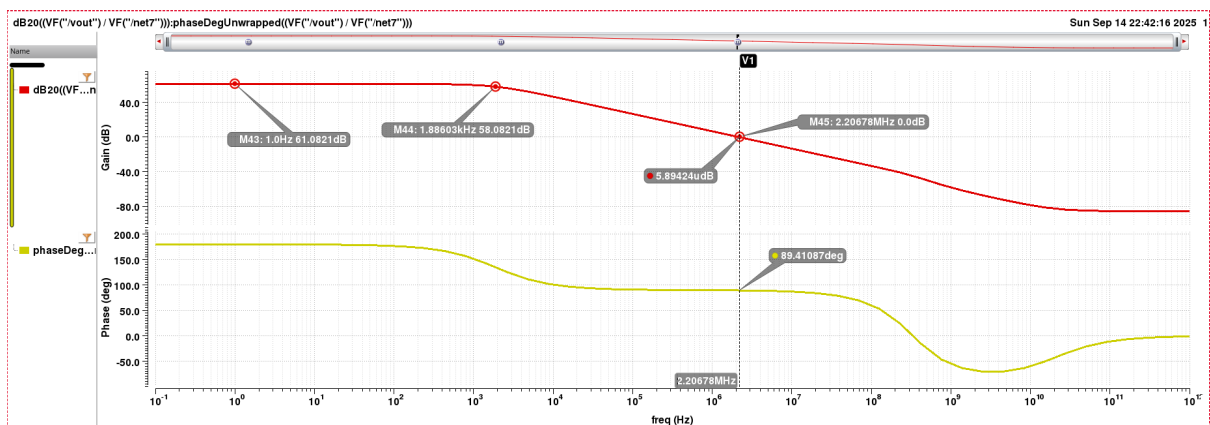


Figure 3: Gain and Phase Plot

Name	Value
1 L8	2.4
2 W8	250m
3 L1	710m
4 L2	710m
5 W1	760m
6 W2	1.51
7 L6	2.4
8 L7	2.4
9 W6	360m
10 W7	250m
11 L0	2.4
12 L4	710m
13 L5	710m
14 W0	576m
15 W4	680m
16 W5	760m

1 dc	☒	t
2 tran	☒	0 10m moderate
3 ac	☒	100m 1T Automatic Start-Stop

Name/Signal/Expr	Value	Plot	Save	Save Options
1 vout		☒	☐	allv
2 vin		☒	☐	allv
3 power	5.71954u	☒	☐	
4 dB20((VF("/vout") / VF("/n...	wave	☒	☐	
5 phaseDegUnwrapped((V...	wave	☒	☐	
6 Gain Mag(dB)	61.0821	☒	☐	
7 Bandwidth	1.88602K	☒	☐	
8 UGB	2.20678M	☒	☐	

Figure 4: Figure of merits

Observations

Length (L)	Width (W)
L1 = 710 nm	W1 = 760 nm
L2 = 710 nm	W2 = 1.51 μ m
L3 = 710 nm	W3 = 360 nm
L4 = 680 nm	W4 = 680 nm
L5 = 710 nm	W5 = 760 nm
L6 = 2.4 μ m	W6 = 2.4 μ m
L7 = 2.4 μ m	W7 = 250 nm
L8 = 2.4 μ m	W8 = 250 nm
L0 = 2.4 μ m	W0 = 576 nm

Table 1: Ordered Transistor Length and Width Values

- **Gain(dB):** 61.0821 dB
- **UGB:** 2.20678 MHz
- **Bandwidth:** 1.88602 kHz
- **Power:** 5.71954 μ W

Question 2

Compare your Assignment 3 common-source (CS) amplifier with the cascode amplifier by sweeping C_L from 20 fF to 1 pF (use parametric analysis).

- For each C_L , record the gain, BW, and GBW.

Briefly explain the observed trends (e.g., pole shift with C_L , impact on GBW and bandwidth).

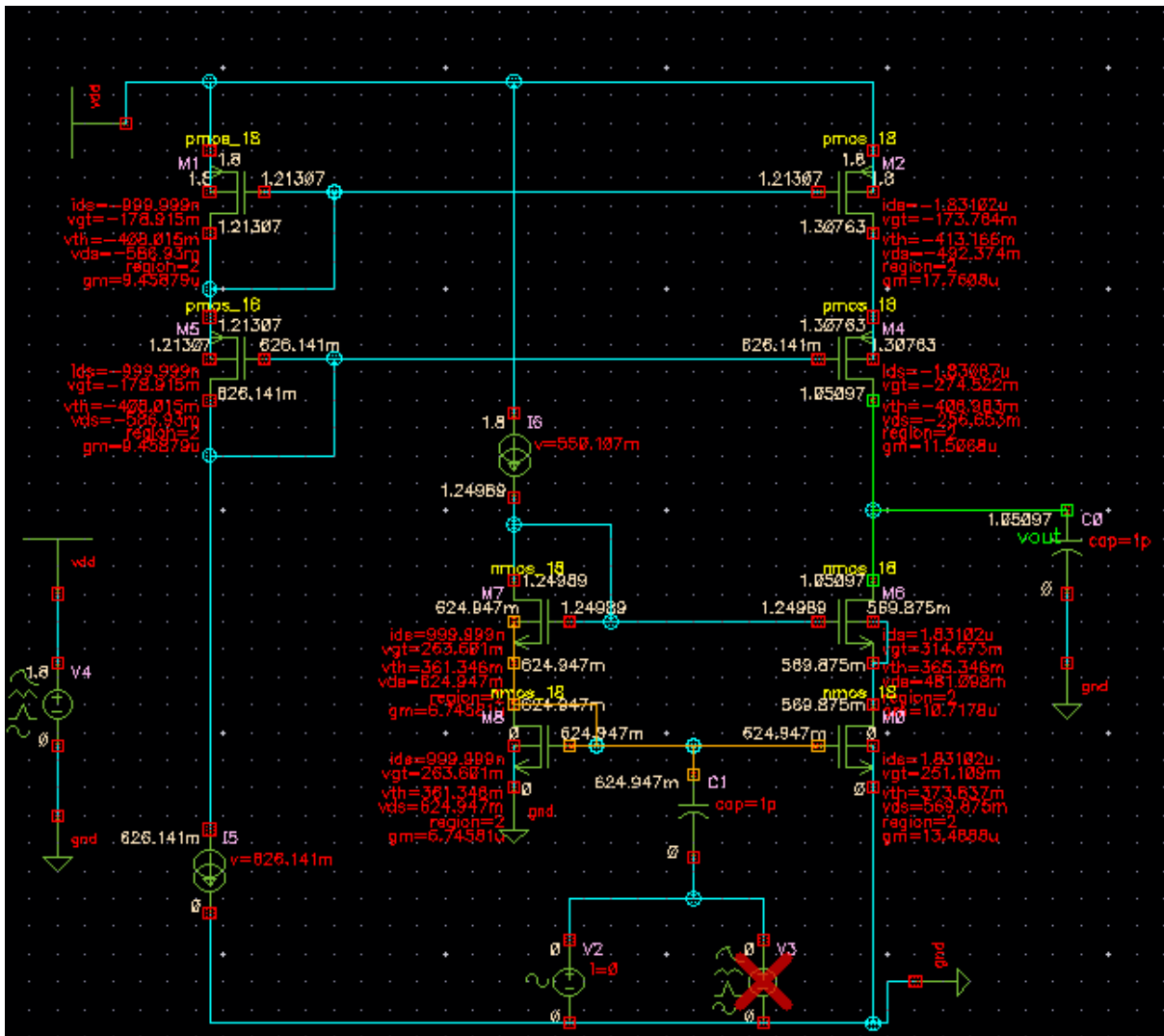


Figure 5: Schematic of Assignment-4 (CS Cascode Amplifier)

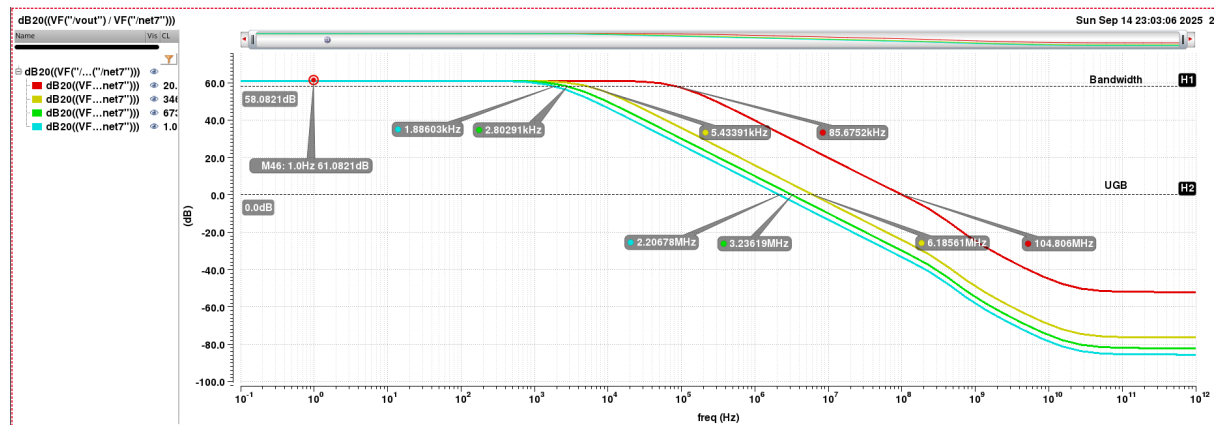


Figure 6: (Parametric Analysis) Changing CL of CS Cascode Amplifier

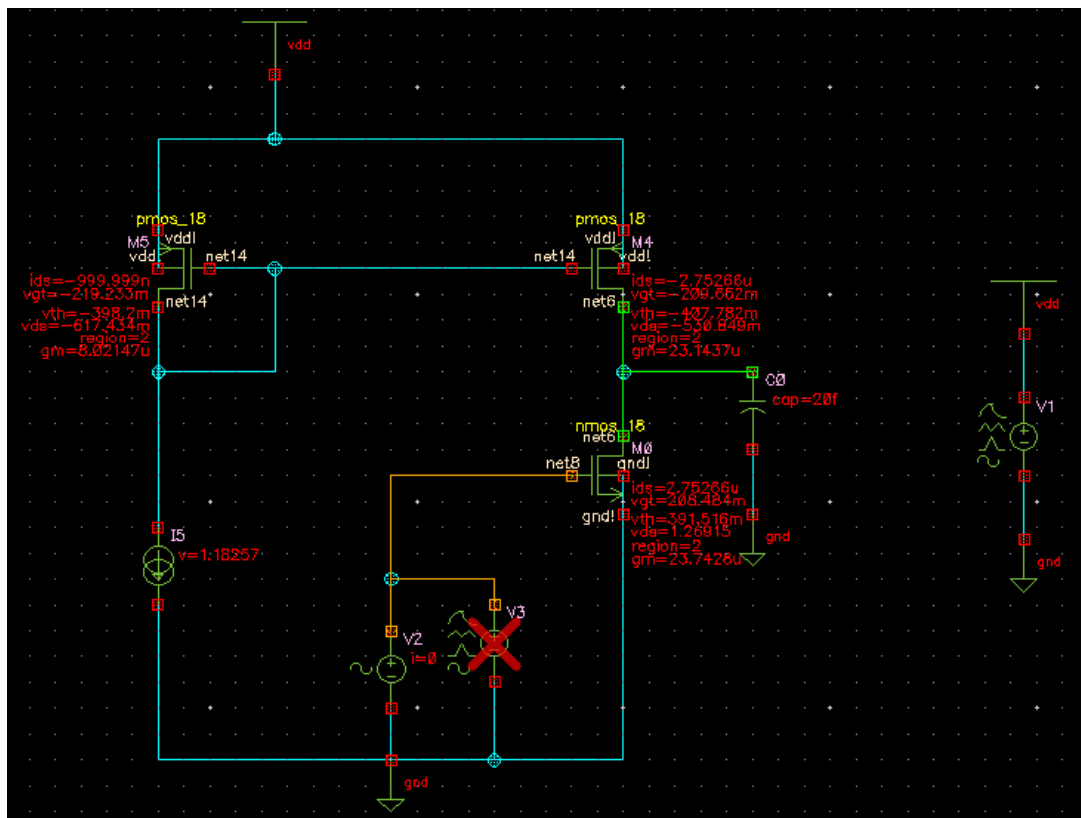


Figure 7: Schematic of Assignment-3 (CS Amplifier)

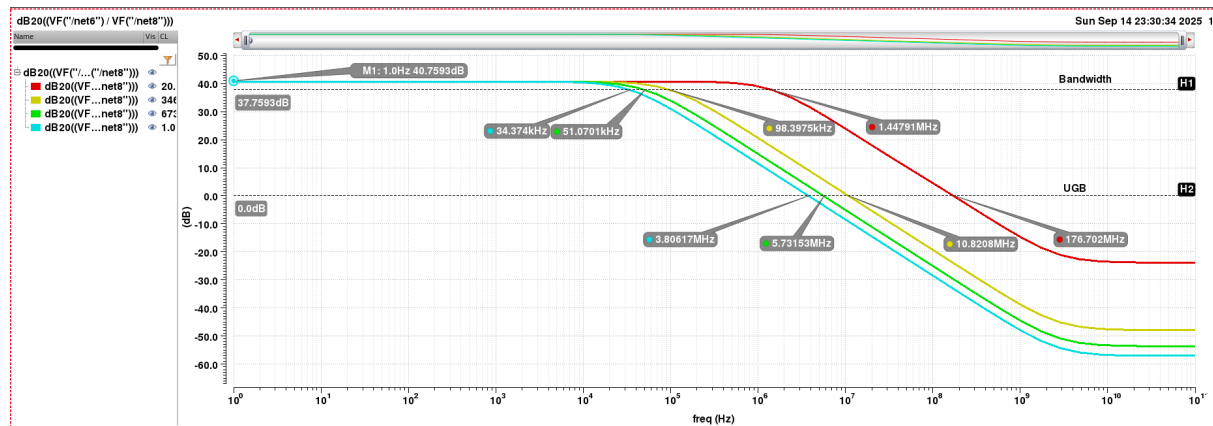


Figure 8: Parametric Analysis) Changing CL of CS Amplifier

Observations

CL (fF)	Bandwidth (Hz)	UGB (Hz)
20	85.68 kHz	104.81 MHz
346	5.43 kHz	6.19 MHz
673	2.80 kHz	3.24 MHz
1000	1.89 kHz	2.21 MHz

Table 2: Bandwidth and UGB for different C_L values (CS Cascode Amplifier)

CL (fF)	Bandwidth (Hz)	UGB (Hz)
20	98.40 kHz	176.70 MHz
346	51.07 kHz	10.82 MHz
673	34.37 kHz	5.73 MHz
1000	1.45 MHz	3.81 MHz

Table 3: Bandwidth and UGB for different C_L values (CS Amplifier)

- As Capacitance increases Bandwidth and UGB both decreases.
- In CS amplifier the bandwidth and UGB are more compared to Cascode Amplifier.